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PSYCHOLOGY'S UNSUNG HERO: HERMANN EBBINGHAUS

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Abstract – Ebbinghaus revolutionized psychology by unlocking the study of higher mental processes. Nonsense syllables created the foundation of the first research in memory, and his meticulous records and preparation opened the door for future studies. Ebbinghaus is best known for his forgetting curve and his discovery of the spacing effect. His work sparked debates in the usefulness of associations and serial learning. Ebbinghaus's contributions also include mapping the history of psychology, being the first to use the literature review, methods, results, discussion format, and he addressed confirmation bias as well as the capacity of short-term memory. Critics of Ebbinghaus frequently employ a hindsight bias that does not align with Ebbinghaus's intentions, or they misrepresent his work.

Keywords: Hermann Ebbinghaus, memory, history of psychology, savings during relearning, associations, spacing effect, forgetting curve

Hermann Ebbinghaus did not believe lack of information on a subject should constitute a lack of effort to experiment with it (Ebbinghaus, 1885/1913). Ebbinghaus (1885/1913) was the earliest pioneer in the study of human memory because he knew that someone had to be the first. His work set the standard for all future experiments on the higher mental processes (Fuchs, 1997), and it confirmed that memory could be studied (Murdock, 1985). Many people know of Ebbinghaus, but very few have taken the time to read his book (Roediger 1985; Slamecka, 1985a). Due to this lack of knowledge, many people do not fully understand what Ebbinghaus accomplished. This is of particular concern when his work is misrepresented and or criticized. Greater knowledge of Ebbinghaus is required to fully understand the gravity of all he discovered and to develop personal opinions on Ebbinghaus's work. After closer inspection, one may conclude that Ebbinghaus is the unsung hero of psychology.

In 1885, Ebbinghaus wrote his most famous book, *Memory: A Contribution to Experimental Psychology*. This work was the compilation of multiple experiments on memory conducted by Ebbinghaus between the years of 1879-1880 and 1883-1884 (Ebbinghaus, 1885/1913). It changed psychology by confirming that studying the higher mental functions was possible (Ebbinghaus, 1885/1913; Fuchs, 1997; Murdock, 1985; Slamecka, 1985b). Ebbinghaus (1885/1913) based his experiments on multiple series of approximately 2,300 nonsense syllables that he created. A nonsense syllable is a three-letter word, that has no meaning, consisting of a consonant-vowel-consonant (Ebbinghaus, 1885/1913). Xat would be an example of a nonsense syllable using this formula, but cat would not. He chose nonsense syllables for his experiments because he did not want interest level to affect the results between different series as would have been the case with poetry (Ebbinghaus, 1885/1913). In addition, nonsense syllables were useful for Ebbinghaus's study because he wanted memory at its simplest (Roediger, 1985) along with the ability to numerically record his results (Ebbinghaus, 1885/1913). Ebbinghaus (1885/1913) made multiple lists of different numbers of nonsense syllables. The number of nonsense syllables he included on a particular list was determined by what he was studying about memory at the time. He proceeded to study these repeatedly until he had faultlessly memorized them and could repeat a list twice consecutively without a mistake (Ebbinghaus, 1885/1913).

Ebbinghaus was so thorough that after initially completing his experiment, in 1880, he replicated it from 1883-1884 (Ebbinghaus, 1885/1913; Fuchs 1997). This was, at least partially, because he wanted to clear his mind of any training that may have occurred before

beginning the second series of experiments (Ebbinghaus, 1885/1913). While the earlier experiment required two correct repetitions, the second rendition only required one repetition without error (Ebbinghaus, 1885/1913). The basic model for most of his experiments was that he would learn a set of nonsense syllables and then document the number of repetitions required later to get back to the point of "first possible reproduction" (Ebbinghaus, 1885/1913, p. 10).

A review of Ebbinghaus's study of memory cannot be complete without an emphasis on how meticulous he was. Ebbinghaus precisely kept track of all the data from his experiments (Ebbinghaus, 1885/1913). This included how long it took to say a nonsense syllable, approximately four-tenths of a second; this mattered because he tracked how many repetitions were required to learn a set by timing himself (Ebbinghaus, 1885/1913). He even tried to keep a consistent tone in repeating the syllables (Ebbinghaus, 1885/1913). When he determined that keeping the same accent was impossible, he decided to purposefully accent every set number of syllables (Ebbinghaus, 1885/1913). Foremost, he cared about experimentation over conjecture (Postman, 1968); therefore, he carefully controlled all variables in his experiment, and his results verified that the study of memory was possible (Ebbinghaus, 1885/1913; Murdock, 1985). Ebbinghaus only wanted to publish refined thoughts based on experiments, and this may have been why his volume of work is not as large as others in the field (Woodworth, 1909).

In *Memory: A Contribution to Experimental Psychology*, Ebbinghaus ran 13 different experiments using himself as the test participant (Ebbinghaus, 1885/1913; Roediger, 1985). In one experiment, Ebbinghaus recited over 15,000 nonsense syllables (Roediger, 1985). Ebbinghaus used his results to measure savings during relearning (Nelson, 1985). This method answers the question of whether what is completely forgotten still exists somewhere in the mind (Ebbinghaus, 1885/1913; Roediger, 1985). To calculate savings during relearning, Ebbinghaus memorized sets of nonsense syllables and then tested how long it took to relearn them after a certain time interval: 20 minutes, one hour, nine hours, one day, two days, six days, and 31 days (Ebbinghaus, 1885/1913). He calculated the amount forgotten by calculating what percentage of the series was remembered and then subtracting that percentage from 100. After displaying the usefulness of his method, Ebbinghaus (1885/1913) proceeded to test many of his questions about memory such as how much a list needs to be studied until it never needs to be reviewed again (Ebbinghaus, 1885/1913). Three major discoveries will be further detailed.

The Forgetting Curve

Ebbinghaus's most famous contribution to psychology is his forgetting curve. Ebbinghaus himself did not graph the forgetting curve (Roediger, 1985), but he supplied the data to make the curve from his 1879-1880 experiments (Ebbinghaus, 1885/1913). Using the savings during relearning method, Ebbinghaus (1885/1913) found that, after 20 minutes, 40% of the list had been forgotten. His lists showed a varying degree of forgetting over time where after one, six, and 31 days about 66%, 75%, and 79% of the lists had been forgotten respectively (Ebbinghaus, 1885/1913).

In fact, after nine hours, the rate of forgetting slowed (Ebbinghaus, 1885/1913). Ebbinghaus's next time gap was 24 hours. From the results of other time intervals less than a day, the number of forgotten items was fewer than Ebbinghaus expected at 24 hours (Ebbinghaus, 1885/1913). Unlike the nine-hour interval, Ebbinghaus (1885/1913) slept between memorization and relearning in the 24-hour period. Even though sleep was a possible explanation for the difference between nine and 24 hours, Ebbinghaus did not accept this hypothesis (Ebbinghaus, 1885/1913). He suggested that others test his results (Ebbinghaus, 1885/1913), and his experiment was repeated with others finding similar irregular intervals between nine and 24 hours (Murre & Dros, 2015). In contrast, Jenkins and Dallenbach (1924) disagreed with Ebbinghaus and created an experiment to test sleep's effect. They confirmed that sleep does slow forgetting (Jenkins & Dallenbach, 1924). Jenkins and Dallenbach (1924) theorized that being awake interferes with old memories because of the continual introduction of new memories.

Associations

The final chapter of Ebbinghaus's book was devoted to associations. Ebbinghaus (1885/1913) articulated and supported the idea that adjacent items on a list will become associated in memory. He decided to test associations further to see whether remote associations, associations between nonadjacent syllables, existed (Ebbinghaus, 1885/1913). His results confirmed the presence of remote associations (Ebbinghaus 1885/1913; Slamecka, 1985a).

Afterward, Ebbinghaus (1885/1913) wondered about the strength of remote associations in comparison to adjacent syllables. He tested many questions such as how different-sized gaps between remote associations affect the ability to relearn newly ordered sets, how reversing a list affects relearning ability with and without a gap, and the effect of repetitions on remote associations

(Ebbinghaus, 1885/1913). He did this by creating a list, learning the list, and then reordering the list to apply to the specific experiment (Ebbinghaus, 1885/1913). Ebbinghaus (1885/1913) was not sure whether he had truly shown the existence of remote associations or whether he had confirmation bias. Therefore, he suggested others test his results (Ebbinghaus, 1885/1913).

His challenge was accepted, and a lively scientific discussion grew out of that last chapter of his book. Ebbinghaus's beliefs were later labeled as part of the chaining hypothesis. This states that neighbors in a series become associated (Murdock, 1985). While the chaining hypothesis is now considered incorrect for memory overall (Postman, 1968; Slamecka, 1985a), its usefulness may still apply to serial learning (Murdock, 1985; Slamecka, 1985a). Slamecka (1985a) admits that more study is still needed on serial learning.

In addition to the chaining hypothesis, many different scientific ideas have arisen on associations. Pavlov's experiments came soon after Ebbinghaus, but not soon enough to have influenced Ebbinghaus. However, research after Ebbinghaus was influenced towards investigating stimulus-response measurements and paired-response learning (Postman, 1968; Nelson, 1985; Slamecka, 1985a). Stimulus-response was not as fruitful as many had hoped regarding serial learning (Slamecka, 1985a). The investigation of stimulus-response in memory was best described by Slamecka (1985a) as "a questionable undertaking" (p. 422). Stimulus-response measures did not translate well from behavioral processes to cognitive processes (Slamecka, 1985a).

Others directly tested Ebbinghaus's experiments on associations. For example, some wondered if Ebbinghaus's original list negatively impacted the newly ordered list of the same syllables. They found that it did (Slamecka, 1985a). Ebbinghaus (1885/1913) thought that he had confirmed and then disconfirmed the number seven as what could be held in short-term memory. However, Slamecka (1985a) argued that because Ebbinghaus's savings during relearning decreased with more remote syllables then the mental confine of short-term memory is not part of the equation. If the capacity of short-term memory was important, then there would not immediately be a difference in forgetting between further associates.

The Spacing Effect

When considering his results, Ebbinghaus made an interesting discovery. In an early experiment, Ebbinghaus (1885/1913) found that the more time spent repeating a list, or the more times it was repeated, the longer it would be remembered. Later, Ebbinghaus experimented with learning sets by heart over a week (Ebbinghaus, 1885/1913). When Ebbinghaus learned sets to perfection, he discovered that relearning those same sets to perfection could be accomplished more quickly a day later (Ebbinghaus, 1885/1913). When he compared the results, he found that it was more efficient to space out the relearning of series.

Today, this is known as the spacing effect, and it is one of Ebbinghaus's largest contributions to psychology (Roediger, 1985). Additionally, he found that increasing repetitions may be damaging to remembering the series the next day (Ebbinghaus, 1885/1913). In a later experiment, Ebbinghaus (1885/1913) attempted to investigate the effect of continued rehearsal after achieving perfection. When compared to learning to the point of the first possible reproduction, he found that quadrupling the number of repetitions, after reaching the first possible reproduction, only increased savings during relearning by about half as much (Ebbinghaus, 1885/1913). Therefore, too much repetition may be detrimental to learning, and spacing out repetitions is better than learning in just one sitting (Ebbinghaus, 1885/1913).

Other Contributions

While the studies noted above are worthy of considerable recognition within psychology, Ebbinghaus does not receive credit for many other ideas he penned in psychology. Ebbinghaus also found that the longer the original series, once learned to perfection, there was less work needed to relearn it (Ebbinghaus, 1885/1913). This could mean that the larger a set of information, the larger return there is for memorizing it. Throughout his monograph, Ebbinghaus is consistently careful of biases. He specifically mentioned trying to eliminate confirmation bias (Ebbinghaus, 1885/1913; Slamecka, 1985a), a bias that would not be labeled for another 92 years by Mynatt et al. (1977). Ebbinghaus should also receive recognition for providing early evidence that the capacity of short-term memory is about seven items (Roediger, 1985). Ebbinghaus discovered this by accident when studying how long it took to learn multiple series of different lengths. He discovered he only needed to repeat a series if it was longer than seven syllables (Ebbinghaus, 1885/1913; Roediger, 1985). Today, psychologists know that short-term memory can hold seven plus or minus two items (Miller, 1956). These findings are worthy of substantial recognition in specific areas of psychology.

Ebbinghaus deserves more respect for his effect on the overarching aspects of psychology. Ebbinghaus (1885/1913) was already identifying the roots of psychology within six years of the founding of Wundt's

lab, what many consider to be the origin of psychology as a distinct field of scientific study (Benjamin, 2019). Ebbinghaus (1885/1913) identified Aristotle as the first psychologist, and he recognized other prior psychologists throughout history such as Helmholtz and Herbart. Therefore, Ebbinghaus laid the foundation for all systems of psychology classes in Greek philosophers, psychophysicists, and philosophers before Wundt.

Not only did Ebbinghaus chart the history of psychology, but he also developed the format for psychological papers (Roediger, 1885). His monograph is now the model for research papers in psychology. He began his book by detailing the problem and why it should be studied (Ebbinghaus, 1885/1913). He continued to explain his method for experimenting with the problem followed by the results. Finally, he discussed his findings (Ebbinghaus, 1885/1913; Roediger, 1985). Today, all psychology researchers follow Ebbinghaus's format. His effect on the field is often understated, but he changed the course of the science.

Critiquing Ebbinghaus

Despite his substantial contribution to the subject, there are many critics of Ebbinghaus and his work. Some criticize his work based on misconceptions. Gilliland (1948) described Ebbinghaus as frequently developing headaches, but Ebbinghaus (1885/1913) only reported headaches in one part of a specific experiment. He did not acquire headaches throughout his research. Others claim that Ebbinghaus only used nonsense material in his monograph (Nelson, 1985). However, Ebbinghaus also used Lord Byron's *Don Juan* for memorization in some experiments (Ebbinghaus, 1885/1913; Nelson, 1985). Both views are misconceptions touted by those claiming to be knowledgeable about Ebbinghaus.

Others criticized techniques or the subject matter of Ebbinghaus's studies. However, many of these criticisms ignore crucial counterpoints. For instance, Gilliland (1948) viewed Ebbinghaus's forgetting curve as too rapid because it only included the use of nonsense syllables. Furthermore, Gilliland (1948) thought that Ebbinghaus should have gone further than just nonsense syllables, but Ebbinghaus was interested in creating a foundation for the study of memory. Nonetheless, Ebbinghaus included results stating that meaningful material probably requires about one-tenth the amount of time as nonsense syllables for memorization (Ebbinghaus, 1885/1913). In addition, others viewed nonsense syllables as a waste of time that directed research toward serial learning and away from more important areas of memory (Kintsch, 1985; Postman 1968). However, nonsense syllables still have some use today for language learning (Postman, 1968).

Gilliland (1948) also thought Ebbinghaus's results were skewed by the length of the series and the introduction of other series during the intervening period before relearning. Slamecka (1985a) agreed with Gilliland that interference between lists could have affected the results, but Slamecka was also critical of Ebbinghaus's method of repeating even learned items. Instead, he suggested that Ebbinghaus should have used a dropout technique where learned syllables were dropped to avoid distorting the results through overlearning (Slamecka, 1985a). Both had valid points, but they forgot to consider that Ebbinghaus was the only test subject. Since Ebbinghaus was the only participant, he could not eliminate interference in the manner that Gilliland (1948) and Slamecka (1985a) suggested. If he had, Ebbinghaus would have never published his research because he would have died before he completed it. Today's memory studies have rectified this problem because they use more than a single participant (Newman & Loftus, 2012).

Slamecka (1985b) further articulated that those taking issue with Ebbinghaus's use of serial learning all wish he had started with something else. For example, Kintsch (1985) criticized Ebbinghaus for theorizing too often and wished Ebbinghaus had started with information processing. Furthermore, Murdock (1985) wanted Ebbinghaus to start with recognition, and Roediger (1985) wished Ebbinghaus had included research on mnemonics. Nonetheless, even though Murdock (1985) claimed that Ebbinghaus should have started with something other than serial learning, he still admitted that it needs more study today. Roediger's (1985) complaint that Ebbinghaus did not include mnemonics and other memorization strategies does not match Ebbinghaus's purpose of having memory at its plainest. In fact, Ebbinghaus (1885/1913) outlawed mnemonics as part of his experiments in his seven rules for memorization. Kintsch's (1985) view that Ebbinghaus theorized too often fails to consider that Ebbinghaus was conducting the first experiments on memory. Further experimentation was the only response to Ebbinghaus's study or it would not have become a field of research.

Postman (1968) criticized Ebbinghaus's savings during relearning. He thought the technique did not capture differences in ease of learning, materials, and chance. Ebbinghaus (1885/1913), however, provided the answer when he wrote that he hoped all other small variables would cancel themselves out. Nelson (1985) came to Ebbinghaus's defense articulating the continued usefulness of savings during relearning as shown by multiple studies.

Conclusion

Ebbinghaus was a forerunner in experimental psychology, and he deserves to be recognized as such (Woodworth, 1909). When 20th-century scholars claimed memory research was a failure, Slamecka (1985a) rebuked them because Ebbinghaus provided evidence that memory could be tested. Ebbinghaus expanded psychology (Postman, 1968). Without him, psychology may have still been focused on the physiological (Shakow, 1930).

It is easy to judge Ebbinghaus for what he did not know (Roediger, 1985; Slamecka, 1985b). Hindsight bias is a danger that many have fallen victim to when reviewing Ebbinghaus's work. However, Ebbinghaus cannot be blamed for the unproductive directions cognitive psychology explored after his death (Slamecka, 1985b). Slamecka (1985b) equated the claim that Ebbinghaus impeded the study of memory to the claim that the Wright brothers hampered the development of the jet. Ebbinghaus made it possible for psychologists to study memory the way they do today. Even if that is all the acknowledgment he receives, Ebbinghaus is still a hero of psychology.

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